

Nama : Irvan Cahya Nugraha

NIM : 245410034

Kelas : IF - 1

RESPONSI SISTEM TERDISTRIBUSI TERDESENTRALISASI SISTEM PENGGAJIAN TUKANG

Aturan masuk (3)									
Cari									
☐	Name	ID aturan grup kea...	Versi IP	Jenis	Protokol	Rentang port	Sumber	Deskripsi	
☐	-	sgr-025c0b46b033d1793	IPv4	SSH	TCP	22	0.0.0.0/0	-	
☐	-	sgr-059627c65c9bbc5d1	IPv4	HTTP	TCP	80	0.0.0.0/0	-	
☐	-	sgr-0cb89b0e345ed72df	IPv4	MYSQL/Aurora	TCP	3306	0.0.0.0/0	-	

Aturan masuk | Aturan keluar | Berbagi | Pengaitan VPC | Tag

Aturan masuk (2)									
Cari									
☐	Name	ID aturan grup kea...	Versi IP	Jenis	Protokol	Rentang port	Sumber	Deskripsi	
☐	-	sgr-063f9135dda0628	IPv4	HTTP	TCP	80	0.0.0.0/0	-	
☐	-	sgr-0c682070afab825be	IPv4	SSH	TCP	22	0.0.0.0/0	-	

```
ubuntu@ip-172-31-20-42:~$ sudo apt install nginx php php-mysql my
l-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
apache2 apache2-bin apache2-data apache2-utils
libapache2-mod-php8.3 libapr1t64 libaprutil1-dbd-sqlite3
libaprutil1-ldap libaprutil1t64 libcgi-fast-perl libcgi-pm-perl
libclone-perl libencode-locale-perl libevent-pthreads-2.1-7t64
libfcgi-bin libfcgi-perl libfcgi0t64 libhtml-parser-perl
libhtml-tagset-perl libhtml-template-perl libhttp-date-perl
libhttp-message-perl libio-html-perl liblua5.4-0
liblwp-mediatypes-perl libmecab2 libprotobuf-lite32t64
libtimedate-perl liburi-perl mecab-ipadic mecab-ipadic-utf8
mecab-utils mysql-client-8.0 mysql-client-core-8.0 mysql-common
mysql-server-8.0 mysql-server-core-8.0 nginx-common php-common
php8.3 php8.3-cli php8.3-common php8.3-mysql php8.3-opcache
php8.3-readline ssl-cert
Suggested packages:
apache2-doc apache2-suexec-pristine | apache2-suexec-custom
www-browser php-pear libdata-dump-perl libipc-sharedcache-perl
libio-compress-brotli-perl libbusiness-isbn-perl
libregexp-ipv6-perl libwww-perl mailx tinycat fcgiwrap nginx-doc
The following NEW packages will be installed:
apache2 apache2-bin apache2-data apache2-utils
libapache2-mod-php8.3 libapr1t64 libaprutil1-dbd-sqlite3
libaprutil1-ldap libaprutil1t64 libcgi-fast-perl libcgi-pm-perl
```

i-04b5e6aeac541d6be (VM-Server)

PublicIPs: 100.31.98.232 PrivateIPs: 172.31.20.42

Pembahasan : Sistem three tier ini (Sistem Penggajian Tukang) saya buat dengan 2 vm yaitu vm-server dan vm-app , pada vm app ada interface untuk user – Tier 1, App script PHP (Logic) – Tier 2 . Pada vm-server ada mysql (Storage) -Tier 3. Konsepnya tukang kasbon ke mandor lalu mandor menginputkan data kasbon dan jumlah kehadiran tukang saat gaji , mempermudah mandor dalam menghitung dan digitalisasi data

VM-App: Izinkan Port 22 (SSH) dan Port 80 (HTTP) dari 0.0.0.0/0.

VM-DB: Izinkan Port 22 (SSH) dan Port 3306 (MySQL)

```
mysql> CREATE USER 'admin_proyek'@'%' IDENTIFIED BY 'Password123!';
Query OK, 0 rows affected (0.03 sec)

mysql> GRANT ALL PRIVILEGES ON sistem_tukang.* TO 'admin_proyek'@'%' ;
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> USE sistem_tukang;
Database changed
mysql>
mysql> CREATE TABLE tukang (
  ->   id INT AUTO_INCREMENT PRIMARY KEY,
  ->   nama VARCHAR(50),
  ->   gaji_per_hari INT
  -> );
Query OK, 0 rows affected (0.05 sec)

mysql>
mysql> CREATE TABLE records (
  ->   id INT AUTO_INCREMENT PRIMARY KEY,
  ->   tukang_id INT,
  ->   kasbon INT DEFAULT 0,
  ->   absen INT DEFAULT 0,
  ->   tanggal DATE,
  ->   FOREIGN KEY (tukang_id) REFERENCES tukang(id)
  -> );
Query OK, 0 rows affected (0.06 sec)

mysql>
mysql> -- Isi data contoh
mysql> INSERT INTO tukang (nama, gaji_per_hari) VALUES ('Budi', 150000), ('Slamet', 140000);
Query OK, 2 rows affected (0.02 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.00 sec)

mysql> EXIT;
Bye
ubuntu@ip-172-31-20-42:~$
```

i-04b5e6aeac541d6be (VM-Server)
PublicIPs: 100.31.98.232 PrivateIPs: 172.31.20.42

Pembahasan : Setting mysql di VM-server membuat Database,Membuat User, Membuat table tukang dan records. Ada data tukang Budi 150000 dan Slamet 140000

```
ubuntu@ip-172-31-25-51:~$ sudo apt update
sudo apt install apache2 php libapache2-mod-php php-mysql -y
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:5 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
0% [4 Packages store 0 B] [6 Translation-en 131 kB/5982 kB 2%] [5
```

i-00badbcf73b150577 (VM-app)



PublicIPs: 3.80.82.197 PrivateIPs: 172.31.25.51

```

GNU nano 7.2 /var/www/html/index.php
<?php
$host = '172.31.20.42'; // GANTI INI PAKAI IP PRIVATE VM 2
$user = 'admin_proyek';
$pass = 'Password123!';
$db = 'sistem_tukang';

$conn = new mysqli($host, $user, $pass, $db);

// Cek Koneksi
if ($conn->connect_error) {
    die("Koneksi Gagal: " . $conn->connect_error);
}

// Logika Simpan & Hitung Gaji (Tier 2 - Application Logic)
if(isset($_POST['hitung'])) {
    $id = $_POST['tukang_id'];
    $kasbon = $_POST['kasbon'];
    $hari = $_POST['hari_kerja'];
    $tgl = date('Y-m-d');

    // Ambil data tarif gaji
    $res = $conn->query("SELECT * FROM tukang WHERE id=$id");
    $row = $res->fetch_assoc();
    $nama = $row['nama'];
    $total_gaji = ($row['gaji_per_hari'] * $hari) - $kasbon;

    // Simpan ke tabel records agar data bisa tampil di bawah
    $conn->query("INSERT INTO records (tukang_id, kasbon, absen, t");

    echo "<div style='padding:10px; background:#d4edda; color:#155724'>
        <h3>Berhasil! Gaji $nama hari ini: Rp " . number_format($total_gaji, 0, '', '.') . "
    </div>";
}
?>

<style>
    body { font-family: sans-serif; margin: 40px; line-height: 1.6}

```

^G Help ^O Write Out ^W Where Is ^R Cut ^T Execute
 ^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

```

GNU nano 7.2 /var/www/html/index.php
    table { width: 100%; border-collapse: collapse; margin-top: 20px; }
    th, td { border: 1px solid #ddd; padding: 10px; text-align: left; }
    th { background-color: #f4f4f4; }
    .form-box { background: #f9f9f9; padding: 20px; border: 1px solid #ddd; }
</style>

<div class="form-box">
    <h2>Sistem Mandor - Input Absensi & Kasbon</h2>
    <form method="post">
        Nama Tukang:
        <select name="tukang_id" required>
            <option value="">-- Pilih Tukang --</option>
            <option value="1">Budi (150rb/hr)</option>
            <option value="2">Slamet (140rb/hr)</option>
        </select><br><br>
        Jumlah Hari Kerja: <input type="number" name="hari_kerja" value="1">
        Jumlah Kasbon (Rp): <input type="number" name="kasbon" value="0">
        <button type="submit" name="hitung" style="padding: 10px 20px;">Hitung</button>
    </form>
</div>

<hr>

<h2>Rekap Data Absensi & Kasbon (Dari Database VM 2)</h2>
<table>
    <thead>
        <tr>
            <th>Tanggal</th>
            <th>Nama Tukang</th>
            <th>Hari Kerja</th>
            <th>Kasbon</th>
            <th>Tarif/Hari</th>
            <th>Gaji Bersih</th>
        </tr>
    </thead>
    <tbody>

```

```

</thead>
<tbody>
  <?php
    $sql = "SELECT t.nama, t.gaji_per_hari, r.kasbon, r.absen,
              FROM records r
              JOIN tukang t ON r.tukang_id = t.id
              ORDER BY r.id DESC";
    $result = $conn->query($sql);

    if ($result->num_rows > 0) {
      while($row = $result->fetch_assoc()) {
        $gaji_bersih = ($row['gaji_per_hari'] * $row['abse
        echo "<tr>
              <td>". $row['tanggal']. "</td>
              <td>". $row['nama']. "</td>
              <td>". $row['absen']. " Hari</td>
              <td>Rp ". number_format($row['kasbon']). "</>
              <td>Rp ". number_format($row['gaji_per_hari]
              <td><strong>Rp ". number_format($gaji_bersi
              </tr>";
      }
    } else {
      echo "<tr><td colspan='6' style='text-align:center;'>B
    }
  ?>
</tbody>
</table>

```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute
 ^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

i-00badbcf73b150577 (VM-app)

PublicIPs: 3.80.82.197 PrivateIPs: 172.31.25.51

Pembahasan : Pada script PHP ada logic untuk menjumlah total gaji setelah dikurangi kasbon dan berapa kali , masih menggunakan system Synchronous

Instans | EC2 | us-east-1 x EC2 Instance Connect | us-east-1 x EC2 Instance Connect | us-east-1 x 3.80.82.197 x +

Not secure 3.80.82.197

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Berhasil! Gaji Slamet hari ini: Rp 830,000

Sistem Mandor - Input Absensi & Kasbon

Nama Tukang:

Jumlah Hari Kerja:

Jumlah Kasbon (Rp):

Rekap Data Absensi & Kasbon (Dari Database VM 2)

Tanggal	Nama Tukang	Hari Kerja	Kasbon	Tarif/Hari	Gaji Bersih
2026-01-14	Slamet	7 Hari	Rp 150,000	Rp 140,000	Rp 830,000
2026-01-14	Budi	10 Hari	Rp 1,150	Rp 150,000	Rp 1,498,850
2026-01-14	Budi	10 Hari	Rp 1,150	Rp 150,000	Rp 1,498,850

Pembahasan : Interface untuk user input data dan melihat data yang ada